

Computer Networks Project
Spring 2025
Computer Networks
CIE 447

Mohamed Magdy 202101520
Mohamed Morshedy 202100438
Ahmed Mohamed 202101027

Supervision

Dr. Mohamed Samir Eid
Eng. Mohamed Saleh
Eng. Amira Ezzeldine



Communications and Information Engineering
Zewail City of Science and Technology

Contents

1	Packet Tracer	3
2	IP Configuration and Subnetting	3
2.1	Subnetting Process	4
3	Connectivity Testing	4
4	Network Changes and Relocation Scenario	7
4.1	5.1 Overview of Network Modifications	7
4.2	5.2 Remote Site Configuration	7
	4.2.1 5.2.1 Implementation Details	7
4.3	5.3 Admin Department Relocation	8
4.4	5.4 Testing the Relocation Scenario	8

1 Packet Tracer

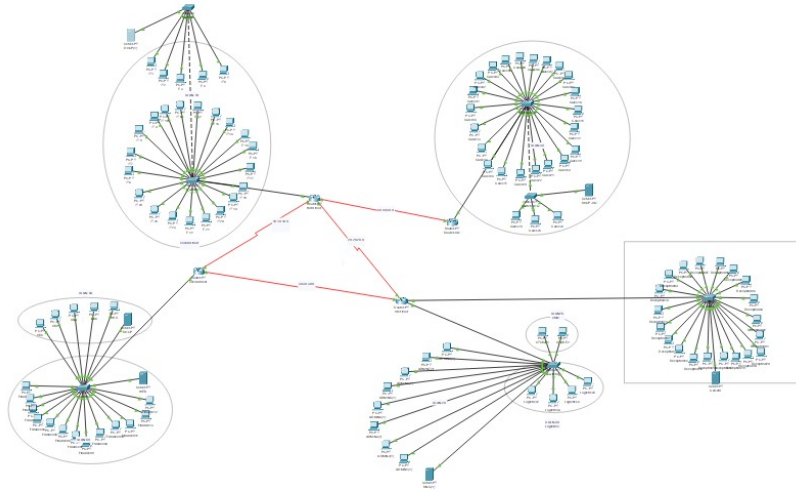


Figure 1: Network Topology

The designed network architecture comprises the following key components:

- **Routers:** Each floor is equipped with a dedicated router, and the routers are interconnected using serial DCE cables to facilitate inter-floor communication.
- **Switches:** Multiple switches are deployed throughout the network to support connectivity across various departments.
- **DHCP Servers:** A Dynamic Host Configuration Protocol (DHCP) server is installed on every floor to automate IP address allocation for the connected devices.
- **DNS Server:** The Domain Name System (DNS) server is hosted within the Administration department and is responsible for resolving domain names within the network.
- **Web Server:** A web server is located in the Finance department and hosts the internal hotel services website at `www.hotelservices.com`.

2 IP Configuration and Subnetting

A subnetting scheme was designed for each department based on the number of connected devices. The IP allocation ensures proper network segmentation using VLANs, maximizing IP address efficiency.

2.1 Subnetting Process

The subnetting strategy was based on the number of hosts required by each department, ensuring optimal IP address allocation with minimal wastage. The corresponding subnet masks were calculated using the formula $2^n - 2$, where n is the number of host bits, accounting for the network and broadcast addresses.

- **IT Department (30 hosts):** Requires 5 bits for host addressing, providing $2^5 - 2 = 30$ usable IP addresses. This corresponds to a subnet mask of /27 (255.255.255.224).
- **Sales Department (25 hosts):** Also needs 5 host bits, yielding a /27 subnet mask.
- **Reception Department (20 hosts):** Similarly requires 5 host bits, resulting in a /27 subnet mask.
- **Finance Department (13 hosts):** Needs 4 host bits, providing $2^4 - 2 = 14$ usable addresses. This maps to a /28 subnet mask (255.255.255.240).
- **Admin Department (9 hosts):** Also requires 4 bits for hosts, resulting in a /28 subnet mask.
- **HR Department (6 hosts):** Needs 3 bits, which yields exactly $2^3 - 2 = 6$ usable IPs. The corresponding subnet mask is /29 (255.255.255.248).
- **Logistics Department (5 hosts):** Similarly needs 3 host bits, resulting in a /29 subnet mask.
- **Store Department (2 hosts):** Requires 2 bits for hosts, giving $2^2 - 2 = 2$ usable addresses. This corresponds to a /30 subnet mask (255.255.255.252).

Department	VLAN	Network IP	Subnet Mask	Starting IP	Ending IP
Reception	80	192.168.2.64/27	255.255.255.224	192.168.2.65	192.168.2.94
Store	70	192.168.2.144/30	255.255.255.252	192.168.2.145	192.168.2.147
Logistics	60	192.168.2.136/29	255.255.255.248	192.168.2.137	192.168.2.142
Finance	50	192.168.2.96/28	255.255.255.240	192.168.2.97	192.168.2.110
HR	40	192.168.2.128/29	255.255.255.248	192.168.2.129	192.168.2.134
Sales	30	192.168.2.32/27	255.255.255.224	192.168.2.33	192.168.2.62
Admin	20	192.168.2.112/28	255.255.255.240	192.168.2.113	192.168.2.126
IT	10	192.168.2.0/27	255.255.255.224	192.168.2.1	192.168.2.30

3 Connectivity Testing

- **Intra-floor Communication:** Devices located on the same floor successfully communicate with one another. For instance, a PC on the second floor was able to ping another PC on the same floor.

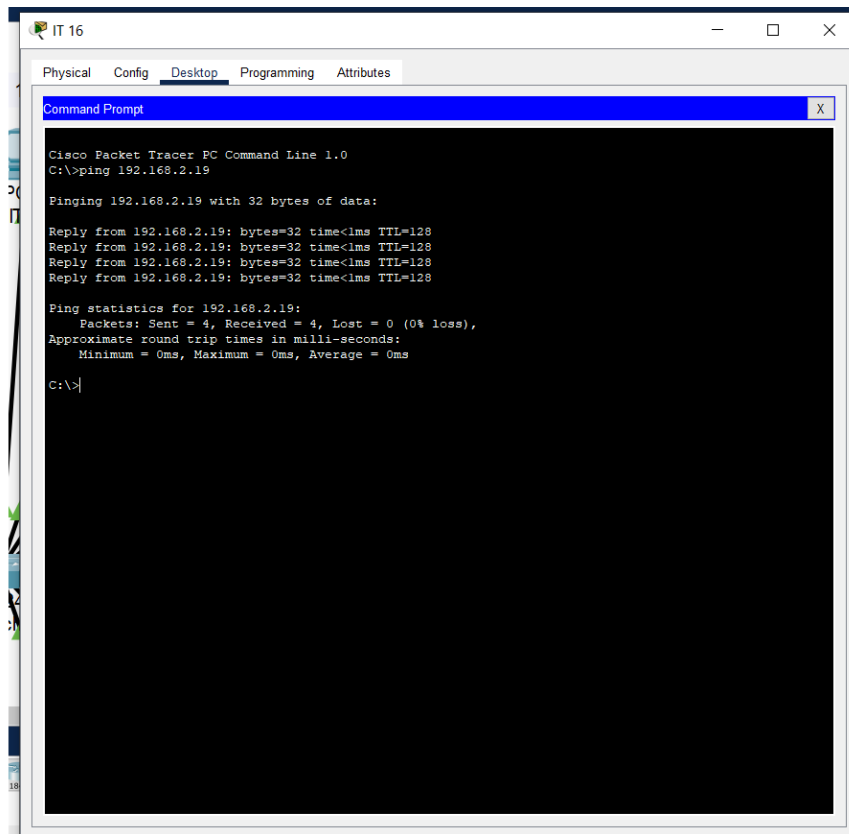
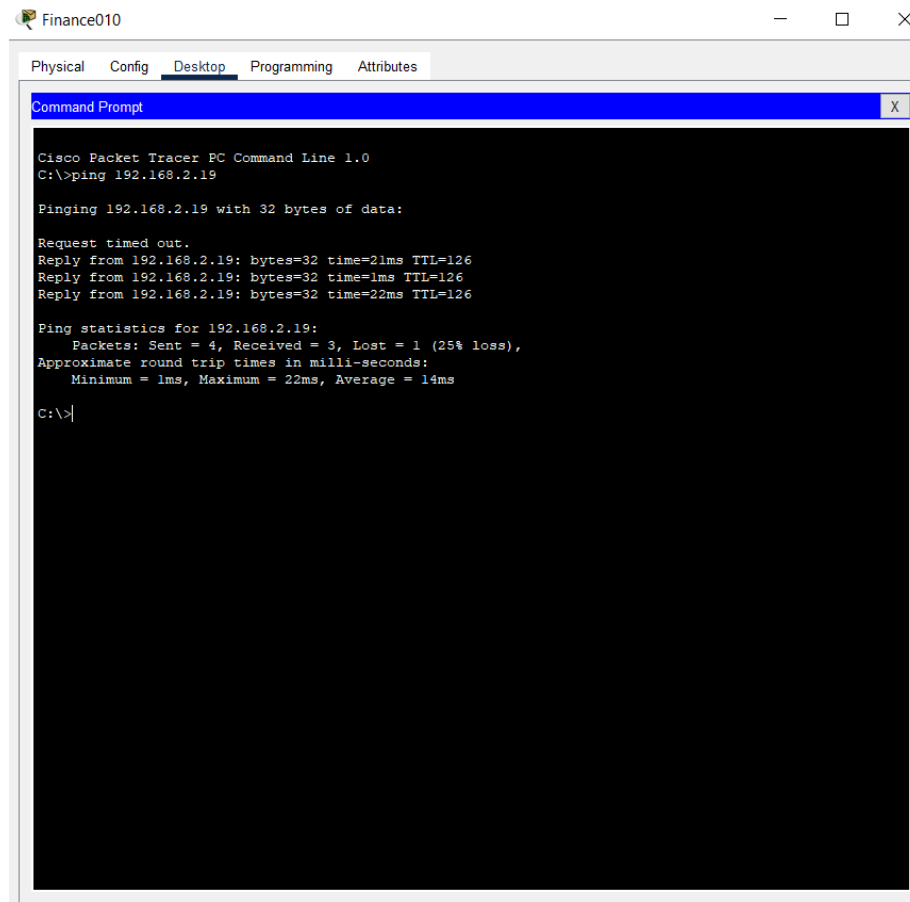


Figure 2: Intra-floor Communication:

- **Inter-floor Communication:** Communication between different floors was validated by sending pings from a second-floor PC to a first-floor PC.



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'Finance010'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. Inside the window, a 'Command Prompt' is open, displaying the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.19

Pinging 192.168.2.19 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.19: bytes=32 time=21ms TTL=126
Reply from 192.168.2.19: bytes=32 time=1ms TTL=126
Reply from 192.168.2.19: bytes=32 time=22ms TTL=126

Ping statistics for 192.168.2.19:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 22ms, Average = 14ms
C:\>
```

Figure 3: Inter-floor Communication:

- **DNS/Web Access:** All devices successfully accessed a locally hosted website on the DNS server, both from within the same floor and in next step across floors.

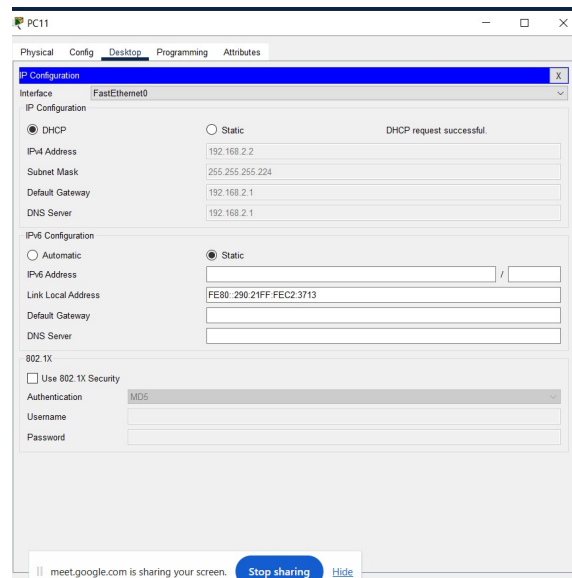


Figure 4:

4 Network Changes and Relocation Scenario

4.1 5.1 Overview of Network Modifications

In response to departmental relocations, several adjustments were made to the hotel network infrastructure:

- The **Sales department** was relocated to the corporate headquarters in another city, requiring remote integration with the hotel network.
- The **Admin department** was moved from the third floor to the first floor of the hotel building.
- Relevant **routing and VLAN configurations** were updated to reflect the new topology and ensure seamless connectivity.

4.2 5.2 Remote Site Configuration

To integrate the relocated Sales department at the headquarters with the main hotel network, the following implementation steps were undertaken:

4.2.1 5.2.1 Implementation Details

- A dedicated router was deployed at the headquarters to manage local connectivity and facilitate communication with the hotel site.

- A Wide Area Network (WAN) link was established between the hotel and headquarters to support inter-site traffic.
- **OSPF (Open Shortest Path First)** routing protocol was configured across the WAN link to enable dynamic route exchange.
- The existing IP addressing scheme for the Sales department was preserved to minimize configuration changes and maintain consistency.

4.3 5.3 Admin Department Relocation

To accommodate the Admin department on the first floor, the following network changes were implemented:

- The first-floor router was reconfigured to include the VLAN and interface settings for the Admin department.
- DHCP services were set up on the first floor to provide dynamic IP address allocation to Admin department devices.
- OSPF routing was updated to reflect the new interface and VLAN locations.
- The previous configuration associated with the Admin department was removed from the third-floor router to avoid conflicts.

4.4 5.4 Testing the Relocation Scenario

To validate the updated network design, comprehensive connectivity testing was performed:

- Devices within the relocated Admin department were successfully pinged from other floors, confirming intra-site communication.
- Devices in the remote Sales department subnet were reachable via the WAN link, verifying inter-site connectivity.
- Routing tables were reviewed and confirmed to contain accurate OSPF-learned routes for all subnets.
- DHCP functionality was tested on the first floor for the Admin department, confirming successful IP lease allocation.

These modifications ensured that both local and remote departments remained fully integrated into the network without loss of service or functionality.